LESSON 4 • MATH IN DAILY LIFE

Multiplication & Division at the Kindergarten



VOCABULARY PART 1

Multiplication Terms & Properties

1. Multiplication

Repeated addition of equal groups.

2. Factor

Number multiplied to get product.

3. Product

Result of multiplying factors.

4. Commutative property

Swapping factor order keeps product.

5. Associative property

Grouping factors keeps product.

6. Distributive property

Multiplying across a sum.

VOCABULARY PART 2

Division & Remainder Terms

7. Division
Splitting into equal parts.

8. Dividend
Total number being divided.

9. Divisor
Number dividing the total.

10. Quotient
Answer from division.

11. Divisible
Divides cleanly with zero left.

12. Division with remainder
Division with items left over.

13. Remainder
Extra amount left undivided.

Arranging Workshop Chairs

Multiplication • Factors • Product • Commutative



David: "By **multiplication**, 3 and 4 are **factors**, giving a **product** of 12 seats!"

Sarah: "Arrange them as 4 × 3 and it's still 12—the **commutative property of multiplication**!"

Folders & Teabreak Trays

Division • Dividend • Divisor • Quotient • Divisible



David: "For folders, (2x3)x4=2x(3x4) by the **associative property of multiplication**!"
Sarah: "And 2 trays of (4+2) pastries equal 2x4+2x2—the **distributive property**!"

Handing Out Workshop Notebooks

Division • Dividend • Divisor • Quotient • Divisible



David: "Let's use **division**! 12 is the **dividend**, 4 teams are the **divisor**, and the **quotient** is 3!"
Sarah: "Clean split! 12 is **divisible** by 4."

Teabreak Coffee Cups

Division with remainder • Remainder



David: "14 cups for 4 teams requires **division with remainder**: 14 ÷ 4 = 3 with 2 left over."

Sarah: "14 is not divisible by 4, so the **remainder** of 2 cups is for us!"

KEY RULES

Three Properties of Multiplication

Commutative Property

$$3 \times 4 = 4 \times 3$$

Swapping factors gives the same product.

Associative Property

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

Regrouping factors gives the same product.

Distributive Property

$$2(4 + 2) = 2{\times}4 + 2{\times}2$$

Multiply each addend inside parentheses.

DIVISION COMPARISON

Divisible vs. Division with Remainder

Workshop Case	Dividend ÷ Divisor	Quotient & Remainder
12 notebooks for 4 teams	**12 ÷ 4**	**Quotient = 3 (Divisible)**
14 coffee cups for 4 teams	**14 ÷ 4**	**Quotient = 3, Remainder = 2**

Fill the Gap Exercise

VOCABULARY PRACTICE

Word Bank: Multiplication, Product, Factor, Division, Quotient, Dividend, Divisor, Divisible, Division with remainder, Remainder, Commutative property of multiplication, Associative property of multiplication, Distributive property

1. To quickly calculate the total number of items arranged in rows and columns (e.g., finding the area or total items), we use the operation of _____.
2. In the expression 252=50, the numbers 25 and 2 play the role of a _____, while 50 is called the _____.

1. When applying the _____, we can change the order or group numbers differently when multiplying multiple terms together without altering the final outcome.
2. To calculate 12(10+3) efficiently by expanding it as (1210)+(123), we rely on the _____.
3. _____, and it must always be strictly less than the divisor.
4. In a standard division setup like 36:12=3, the number 36 being shared is the _____, and 12 is the _____.

Fill the Gap Exercise

VOCABULARY PRACTICE

Word Bank: Multiplication, Product, Factor, Division, Quotient, Dividend, Divisor, Divisible, Division with remainder, Remainder, Commutative property of multiplication, Associative property of multiplication, Distributive property

1. When a natural number can be divided by another with no left-over amount, we say that the first number is _____ by the second.
2. If we distribute 36 items among 15 people and find that some are left over, we are performing a _____.
3. In the division statement 36:15=2(rem.6), the amount left over (6) is specifically called the _____, and it must always be strictly less than the divisor.

WORKSHOP
READY!

You Mastered Kindergarten Math!

Multiplication • Product • Factor • Division • Quotient • Dividend • Divisor • Divisible • Remainder • Commutative • Associative • Distributive